Hernán Salinas Castro

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Interests

PhD student in Astronomy with experience in exoplanet characterization through the transit method and Markov Chain Monte Carlo (MCMC) techniques. Currently focused on the study of the chemical composition of protoplanetary disks using observations from the James Webb Space Telescope (JWST). My goal is to advance our understanding of the chemistry of exoplanetary atmospheres and protoplanetary environments to better constrain planet formation processes.

Education

Pontificia Universidad Católica de Chile — Santiago, Chile
PhD in astrophysics — 2026 - Present

Universidad Andrés Bello — Santiago, Chile
Bachelor's degree in astronomy — 2020 - 2024

Research experience

Chemistry of FU Orionis Objects with JWST — March 2026-Present

Investigation of FU Orionis (FUor) disks, focusing on the chemical composition of inner disk regions following accretion outbursts using JWST data under the supervision Dr. Viviana Guzmán.

Visitor at ESO (Santiago) — June 2025

Study of exoplanetary atmospheres through high-resolution spectroscopy techniques. Conducted calibration and data analysis of observations obtained with the HARPS and NIRPS spectrographs, focusing on a hot Jupiter-type exoplanet in a close-in orbit around its host star. This work is carried out under the supervision of Dr. Elyar Sedaghati.

VLTI Observations of the Atmosphere of Normal Red Giants — January 2025 - June 2025

Used AMBER data from the VLTI to study red giant stars through optical and infrared interferometry techniques. During this project, I had the opportunity to learn how to calibrate and analyze interferometric data under the supervision of Dr. Keiichi Ohnaka.

Member of the MOTE Project | Monitoring of Transiting Exoplanets (MOTE) Project — December 2023 – Present

Participated in a project focused on training undergraduate students in the analysis of exoplanet transit data. I contributed to the selection and planning of observation nights for target exoplanets using different telescopes, such as SPECULOOS and LCOGT. I was also involved in the calibration and reduction of the collected data, followed by photometric analysis. As part of this work, I developed a sub-project focused on the study of Jupiter-type exoplanets orbiting M-dwarf stars, which became the topic of my undergraduate thesis. Additionally, I had the opportunity to write an observation proposal during my undergraduate studies as part of this sub-project. This entire project was carried out under the supervision of Dr. Claudio Cáceres.

Workshops and courses

IFA-Universidad de valparaíso — Valparaiso, Chile
ALMA Cycle 13 Proposal Preparation — March 20, 2026

NASA Exoplanet Science Institute — Virtual
Silver Jubilee: Exoplanet Demographics — July 21-25, 2025

European Interferometry Initiative (EII) — Virtual
Horizons for Optical Long Baseline Interferometry — January 27-30, 2025

Universidad Andrés Bello — Santiago, Chile
Radio astronomy course — August - December, 2024

Universidad Andrés Bello — Santiago, Chile
Complementary cosmological simulations course — August - December, 2024

Space Mood — Virtual
Space Biology Networking in Peru (BEP) — December 14, 2024

Universidad Andrés Bello — Santiago, Chile
First Astrochemistry Workshop — October 28, 2024

NASA Exoplanet Science Institute — Virtual
Advances in Direct Imaging: From Young Jupiters to Habitable Earths — July 22-26, 2024

24RED — Virtual
The virtual astrobiology introductory course — June 2024

Centro de Astrofísica y Tecnologías Afines (CATA) — Santiago, Chile
CATA Area 6 (Exoplanetas y astrobiología) Meeting — May 7, 2024

Universidad Andrés Bello — Santiago, Chile
Complementary interferometry course — March - June 2024

Talks

Centro de Astrofísica y Tecnologías Afines (CATA) Santiago, Chile
CATA Area 6 Meeting August 6, 2025
"Jupiter type exoplanets orbiting M dwarf stars"

Outreach activities

Member of the Astroamigos Wintata Wara Group April 2025 - Present

A astronomy group engaged in public science outreach through activities such as telescope observations, public talks, interactive stations, and workshops. Within the group, I serve as a science communicator and collaborator, supporting content development and assisting with telescope operations. I have participated in events at institutions such as Observatorio Calán, Observatorio Manuel Foster, and Museo Nacional Aeronáutico y del Espacio. Instagram:@astroamigos_wintata_wara.

Galaxies for Children Workshop May 2023

Together with two colleagues, I organized an outreach workshop for children at Luis Cruz Martínez School in Andacollo, Chile. We introduced basic concepts about galaxies and guided an interactive activity where the children created their own galaxies using plasticine. The goal was to foster creativity, curiosity, and interest in astronomy.

Technical Skills

Languages: Spanish - Intermediate English.
Programming language: Python and LaTeX(intermediate level).
Bash script (Basic level).
Sofware:
Intermediate level: Topcat, Aladin, SAOImageDS9.
Basic level: Excel, Iraf, Astroimagej, Casa, Carta, FV.
TESS telescope data analysis(Basic level).
Photometric data analysis for exoplanets:
Intermediate level: exotic_ld, ccdproc, batman, lightkurve, emcee.